# Quiz — 1.3.1 Compression, Encryption and Hashing — ANSWER SHEET

*Separate answer sheet / mark scheme — keep for marking.*

## Answer key

**Section A (8 marks, 1 each)**

1. **B** — Lossy permanently discards data; the original cannot be exactly recovered.
2. **B** — RLE replaces runs of repeated values, so long runs compress well.
3. **A** — With few/short runs, the value+count pairs can exceed the original size.
4. **C** — The dictionary mapping patterns to references must accompany the file to decode it.
5. **B** — Symmetric uses one shared key for both encryption and decryption.
6. **B** — Symmetric's key-distribution problem is solved by the openly shared public key.
7. **B** — Sign with private key, verify with public key (proves origin/integrity).
8. **C** — Deterministic: same input always yields the same digest.

**Section B (12 marks)**

9. **[2]** 1 mark: no data is permanently removed / original is exactly recoverable. 1 mark: any valid use, e.g. text/program code/ZIP/PNG.

10. **[2]** Encoded output: `5W 2B 6W` (or `W5 B2 W6`). 1 mark correct values W,B,W in order; 1 mark correct counts 5,2,6.

11. **[2]** Decoded: `AAAAA BBB C AAAA` i.e. `AAAAABBBCAAAA`. 1 mark for correct letters in order; 1 mark for correct counts (5,3,1,4).

12. **[2]** 1 mark: the public key can be shared openly so no secret key needs to be exchanged in advance. 1 mark: only the matching private key (kept secret) can decrypt, so an intercepted public key/ciphertext is useless — avoids the symmetric key-distribution problem.

13. **[2]** Any two for 1 mark each: hashing is one-way so the original password cannot be retrieved even if the database is stolen; the system only compares hashes at login so it never needs to store/recover the plaintext; encryption is reversible with a key (a risk if the key is compromised).

14. **[2]** Any two for 1 mark each: one-way/not reversible; deterministic; low/few collisions; fast to compute; produces fixed-length output.

**Section C (5 marks — award up to 5 from)**

15.
   - Compression reduces audio file size so it transfers faster and uses less bandwidth/storage. **[1]**
   - Lossy compression (e.g. MP3) is suitable for audio because slight quality loss is acceptable and gives a much smaller file. **[1]**
   - Encryption scrambles data in transit so an interceptor cannot read it; (a)symmetric encryption is used — HTTPS uses asymmetric to exchange a symmetric session key, then symmetric for the bulk audio stream. **[1]**
   - Hashing the password means only the hash is stored, not the plaintext. **[1]**
   - At login the entered password is hashed and compared; because hashing is one-way a stolen database does not reveal the actual passwords. **[1]**

**Total: 8 + 12 + 5 = 25 marks**